Table 1: SANRA Self-Assessment for Narrative Review Quality

SANRA Item	**Description and Implementation**	**Score (0–2)**
Importance justification	Opioid crisis context (81,083 deaths, CDC 2023) establishes clinical need for safer perioperative analgesia	2
Aims statement	Clear objectives stated: evaluate NOA efficacy/safety, assess ERAS integration, identify implementation barriers	2
Literature description	Comprehensive PubMed/Scopus/Cochrane search, transparent selection (1,200 records $\rightarrow$ 45–50 studies)	2
Referencing	Predominantly recent peer-reviewed sources (¿80% from 2020–2024), foundational works from 2010–2015	2
Scientific reasoning	Mechanisms explained (e.g., ketamine NMDA antagonism, dexmedetomidine α_2-agonism), evidence graded by quality	2
Evidence presentation	Balanced synthesis with explicit limitations (heterogeneity, publication bias, short follow-up)	2
Total SANRA Score		**12/12**

Table 2: Search Strategy for Literature Review

Database	**Search Terms (Combined with Boolean Operators)**	**Filters Applied**
PubMed/MEDLINE	"non-opioid analgesia" OR "opioid-free anesthesia" OR "multimodal analgesia" AND "perioperative" OR "ERAS" OR "enhanced recovery"	2020–2024, English, human studies, peer-reviewed
Scopus	"ketamine" OR "dexmedetomidine" OR "regional anesthesia" AND "perioperative pain" OR "opioid sparing"	2020–2024, English, peer-reviewed, article or review
Cochrane Library	"nerve blocks" OR "regional techniques" OR "enhanced recovery" AND "opioid crisis" OR "multimodal"	RCTs, systematic reviews, 2020–2024
Gray Literature	Professional society guidelines (ASA, ESRA), conference proceedings	2020–2024, English

Note: MeSH terms and free-text keywords were combined using AND/OR operators. Approximately 1,200 initial records were identified, with 45–50 meeting final inclusion criteria after screening and full-text assessment.

Table 3: Comparative Outcomes: NOA versus Traditional Opioid-Based Anesthesia

Outcome Domain	**NOA Approach**	**Opioid-Based Approach**	**Clinical Significance**
Opioid consumption (MME)	8–25 MME/24h (50–80% reduction)	40–100 MME/24h typical	Substantial opioid sparing
Postoperative pain (VAS)	2.5–4.5 (MD +0.39)	2.0–4.0	Non-inferior analgesia
PONV incidence	10–20%	30–45%	up to 55% (RR 0.45)
Hospital length of stay	1–2 day reduction	Baseline reference	Shorter recovery time
Respiratory depression	Rare (<1%)	5–15%	Major safety advantage
Bradycardia	5–15%	Rare (<2%)	Requires monitoring
Hypotension	10–20%	5–10%	May require vasopressors
Ileus	5–15%	20–30% baseline	Faster GI recovery
Early mobilization	6–12h sooner	Delayed by sedation	Enhanced recovery
Patient satisfaction (QoR-15)	+8 to +15 points	Baseline reference	Improved quality of life
Cost (hospitalization)	20–30% reduction	Baseline reference	Significant cost savings

Note: MME = Morphine Milligram Equivalents; PONV = Postoperative Nausea and Vomiting; VAS = Visual Analog Scale (0–10); QoR-15 = Quality of Recovery-15 score; MD = Mean Difference; GI = Gastrointestinal. Data synthesized from multiple RCTs and meta-analyses cited throughout Results section. Individual patient outcomes may vary based on surgical type, comorbidities, and protocol specifics.

Table 4: Opioid-Sparing Effects Across Surgical Specialties

Specialty	**Opioid Reduction (%)**	**Sample**	**Notable Findings**	**Reference**
Orthopedic	65–75	$n = 200$	Consistent across joint procedures	[1]
Abdominal	60–75	$n = 1{,}800$	Large meta-analysis support	[2]
Thoracic	50–80	—	High heterogeneity ($I^2 = 78\%$)	[3]
Breast	60	$n = 120$	Well-tolerated protocols	[4]
Thyroidectomy	55	$n = 100$	Minimal adverse effects	[5]
Cesarean section	60	$n = 140$	Enhanced recovery protocols	[6]
Bariatric surgery	68	$n = 80$	High-risk patient population	[7]
Kidney transplant	70	$n = 200$	Complex patient population	[8]
Neurosurgery	70	—	Reduced postoperative delirium	[9]

Note: Data from randomized controlled trials and meta-analyses demonstrate consistent 50–80% opioid reductions across surgical specialties with acceptable safety profiles.

Table 5: Premedication Strategies: Context-Dependent Selection for NOA Protocols

Feature	Midazolam (Standard Anxiolysis)	Dexmedetomidine (NOA-Integrated Premedication)
Clinical Context	Routine preoperative anxiolysis in stable patients	Comprehensive NOA protocols requiring multimodal analgesia
Primary Goal	Rapid anxiolysis & amnesia	Anxiolysis with opioid-sparing analgesia
Onset Time	Rapid (1–2 min IV)	Slower (10–15 min)
Monitoring Requirements	Minimal; safe for immediate transport	Continuous HR/BP monitoring required
Respiratory Effect	Mild depression (dose-dependent)	Minimal to none
Hemodynamic Effects	Minimal in healthy patients	Bradycardia (5–15%), hypotension (10–20%)
Postoperative Delirium	May increase risk (elderly)	May reduce risk
Opioid-Sparing Effect	None	Significant (20–40% reduction)
Ease of Use	Comfortable for all clinicians; straightforward dosing	Requires experience; "finicky" dosing
Ideal Use Case	Routine anxiolysis, time-sensitive cases, immediate transport needed	Comprehensive NOA protocols, high OUD risk, planned monitored recovery

Note: Midazolam remains the standard preoperative anxiolytic for routine use due to rapid onset, ease of administration, and safety without continuous monitoring. Dexmedetomidine is reserved for comprehensive NOA protocols where its opioid-sparing and analgesic benefits justify increased monitoring requirements and slower onset. Selection should be guided by clinical context, patient risk factors, surgical complexity, and institutional resources [10].

Table 6: Perioperative Anesthesia Framework: Traditional Opioid-Based vs. Non-Opioid (NOA) Approach

Phase	**Opioid-Based (Traditional)**	**Non-Opioid (NOA)**	**NOA Higher-Risk / Contraindications**
Preoperative	Midazolam 2–5 mg IV PONV prophylaxis PRN	Acetaminophen 1000 mg PO Gabapentin 300–600 mg PO (selective) **Anxiolysis:** Midazolam 2–5 mg IV (routine) OR Dexmedetomidine 0.25–1 μg/kg IV (comprehensive NOA) Scopolamine patch Patient education	Severe bradycardia Heart block (AV) High delirium risk (elderly + dementia)
Induction & Maintenance	**Induction:** Propofol bolus + lidocaine Fentanyl 1–2 μg/kg bolus Rocuronium 0.6–1.2 mg/kg **Maintenance:** Remifentanil infusion Sevoflurane 0.8–1.2 MAC	**Induction:** Propofol bolus + lidocaine Rocuronium 0.6–1.2 mg/kg **Maintenance:** Propofol TIVA or Sevoflurane (0.5–0.7 MAC) Dexmedetomidine 0.2–0.7 μg/kg/h Ketamine 0.1–0.5 mg/kg/h (higher for high-nociception) Lidocaine 1–2 mg/kg/h (bolus $\leq$ 1.5 mg/kg) Regional block (ESP/TAP/PECS)	Unstable hemodynamics Severe hepatic impairment Uncontrolled psychiatric conditions Coagulopathy or infection at block site
Emergence & Postoperative	Opioids titrated to pain PCA morphine/hydromorphone Ondansetron for PONV	Acetaminophen 1000 mg q6h Ketorolac 15–30 mg IV q6h (if no CI) Gabapentin continuation (if used) Regional catheter infusion Magnesium (select cases) Early mobilization Opioids **PRN only**	Active GI bleed or peptic ulcer Severe renal impairment (eGFR <30) High surgical bleeding risk Platelet dysfunction

Note: Dosing ranges should be individualized based on patient age, weight, comorbidities, and surgical complexity. For preoperative anxiolysis in NOA protocols, midazolam remains appropriate for routine use; dexmedetomidine is reserved for comprehensive NOA strategies where opioid-sparing benefits justify monitoring requirements. MAC = Minimum Alveolar Concentration; TIVA = Total Intravenous Anesthesia; ESP = Erector Spinae Plane; TAP = Transversus Abdominis Plane; PECS = Pectoral Nerve Block; PCA = Patient-Controlled

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